

AI-Assisted Interior Furniture Layout System

- Course Project

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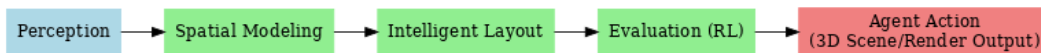
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1. Motivation (Why it Matters)

- Professional relevance: Interior designers spend significant time iterating layouts; automation can reduce repetitive work.
- Everyday relevance: Families face uncertainty when buying furniture (fit, aesthetics); a preview system reduces waste.
- Academic relevance: This project applies AI, optimization, and HCI principles to a real-world design challenge.

2. System Architecture (Three Layers, Five Modules)

The system follows a three-layer architecture, subdivided into five modules.



Three Layers:

- Perception → Process → Action

Five Modules:

1. Perception (input capture, object/furniture detection)
2. State Modeling (2D/3D spatial representation)
3. Intelligent Layout (constraint-based arrangement, heuristics)
4. Evaluation (spatial utilization, movement flow, aesthetics)
5. Agent Action (interactive visualization & user feedback)

3. Scope

In Scope (Core):

- 2D floor plan input
- Basic furniture placement (drag/drop, no overlap)
- Simple evaluation (space utilization, corridor width)
- Simple interface for visualization

Out of Scope (Not included in course project):

- Full CV recognition of furniture and room dimensions
- Complex optimization algorithms (reinforcement learning, multi-objective optimization)
- Commercial integration (e-commerce, dynamic furniture databases)
- AR/VR integration

Optional Extensions (if time allows):

- Simplified 3D visualization
- Automated heuristic-based layout generation
- Extended evaluation metrics

4. Timeline (Aligned with Course Schedule)

Dates	Activities	Deliverables
Sep 5 – Sep 26	Exploration & preparation: select project topic, requirements analysis, tools selection, draft scope & architecture	Draft scope & architecture notes
Sep 29 – Oct 3	Finalize proposal	Project Proposal (Due Oct 3): Motivation, Architecture, Scope, Timeline
Oct 6 – Oct 24	Prototype Phase I: implement 2D modeling & basic furniture placement	Early prototype demo;
Oct 27 – Oct 31	Refine prototype with evaluation features	Progress Report (Due Oct 31): partial prototype with layout + evaluation
Nov 3 – Nov 21	Prototype Phase II: add evaluation rules (no overlap, corridor width); optional heuristic layout	Extended prototype features
Nov 24 – Nov 28	Refinement: improve usability, robustness; optional RL component	Refined prototype
Dec 1 – Dec 5	Final integration & polish; prepare presentation & report	Final draft of report & slides
Dec 8	Present and submit final work	Final Report & Presentation (Due Dec 8)

5. Evaluation Criteria (How Success Will Be Measured)

- **Functionality:** Does the prototype allow furniture placement without overlaps and display layouts?
- **Usability:** Is the interface simple and intuitive for non-experts?
- **Robustness:** Can it handle different room sizes and multiple furniture items?
- **Demonstration Value:** Does it clearly demonstrate AI potential in interior design, even at prototype level?